INCH-POUND

MIL-DTL-32375B (MR)
w/Amendment 3
15 January 2021
SUPERSEDING
MIL-DTL-32375B (MR)
w/Amendment 2
2 July 2019

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY, 7085, 7056, 7075, and 7181, UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers AA7085 and AA7056 wrought aluminum alloy armor plate for non-fusion welded applications in nominal thicknesses from 0.500 to 3.000 inch, inclusive, AA7075 wrought aluminum alloy armor plate for non-fusion welded applications in nominal thicknesses from 1.000 to 1.500 inch, inclusive, and AA7181 in nominal thicknesses from 1.501 to 3.000 inch, inclusive (see 6.2). Additional applications for the AA7056 alloy with a T741 temper is also included in this specification for ballistic plate with an ordered thickness of 2.500 to 3.000 inches. The fusion weldability of these wrought aluminum alloy armors has not been tested or analyzed over this nominal thickness range. Therefore, these alloys should only be used as appliqué armor until a weldability study has been completed with acceptable results.

1.2 Classification. The wrought aluminum armor should be of the following class and tempers (types), as specified (see 6.2).

1.2.1 Class I. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7085 aluminum alloy. The applicable gauge range for class I is 0.500-3.000 inches.

1.2.1.1 Type A. Type A has a temper designation (See 6.4) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.1.2 Type B. Type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

1.2.2 Class II. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7056 aluminum alloy. The applicable gauge range for class II is 0.500-3.000 inches.

1.2.2.1 Type A. Type A has a temper designation (See 6.4) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles.

1.2.2.1.1 Type A1. Type A1 has a temper designation (See 6.4) for special applications in the ordered thickness of 2.500 to 3.000 inches.

1.2.2.2 Type B. Type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

1.2.3 Class III. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7075 aluminum alloy. The applicable gauge range for class III is 1.000 - 1.500 inches.

1.2.4 Class IV. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7181 aluminum alloy. The applicable gauge range for class IV is 1.501 – 3.000 inches. Class IV may be used as a substitute for Class I, Type A or Class II, Type A or A1 when approved by the procuring engineering activity.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V_{50} Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

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THE ALUMINUM ASSOCIATION, INC.

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products
Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products Spectrometry (Performance Based Method)
ASTM E716	-	Standard Practices for Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission

(Copies of these documents are available from https://www.astm.org.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

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3.2 Chemical composition. The chemical composition of the plates shall be within the limits shown in table I. The limits specified in table I were listed from the Aluminum Association Teal Sheets, International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal transverse (LT) direction shall meet the minimum mechanical properties listed in table IIa, table IIb, table IIc, and/or table IId. If mechanical property requirements differ from those contained in table IIa, table IIb, table IIc, or table IId or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE I. Chemical composition, weight percent. 1/ 2/

ELEMENTS	SYMBOL	Class I 7085 ALLOY	Class II 7056 ALLOY	Class III 7075 ALLOY	Class IV 7181 ALLOY
Silicon	Si	0.06	0.10	0.4	0.08
Iron	Fe	0.08	0.12	0.5	0.10
Copper	Cu	1.3 - 2.0	1.2 - 1.9	1.2 - 2.0	1.2 – 1.9
Manganese	Mn	0.04	0.20	0.3	0.15
Magnesium	Mg	1.2 - 1.8	1.5 - 2.3	2.1 - 2.9	1.7 - 2.2
Chromium	Cr	0.04	---	0.18 - 0.28	0.04
Zinc	Zn	7.0 - 8.0	8.5 - 9.7	5.1 - 6.1	6.7 – 7.9
Titanium	Ti	0.06	0.08	0.2	0.06
Zirconium	Zr	0.08 - 0.15	0.05 - 0.15	---	0.08 - 0.18
Other, max. Each	---	0.05	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder	Remainder

1/ Except for “Aluminum” and “others”, analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those “others” metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

TABLE IIa. Minimum mechanical properties for class I, AA7085. 1/, 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Type A	Type B	Type A	Type B	Type A	Type B
0.500 to 1.500, incl.	80	68	74	60	11	12
1.501 to 2.000, incl.	78	67	73	59	11	12
2.001 to 3.000, incl.	77	67	72	58	10	11

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch.
2/ Values are taken in the Longitudinal Transverse Direction (LT).

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TABLE IIb. Minimum mechanical properties for class II, AA7056. 1/, 2/

Thickness, inches	Tensile Strength, ksi			Yield Strength, 0.2% Offset, ksi			Elongation percent		
	Type A	Type A1	Type B	Type A	Type A1	Type B	Type A	Type A1	Type B
0.500 to 0.999, incl.	TBD 3/	N/A 4/	TBD 3/	TBD 3/	N/A 4/	TBD 3/	TBD 3/	N/A 4/	TBD 3/
1.000 to 2.000, incl.	80	N/A 4/	64	77	N/A 4/	56	7	N/A 4/	11
2.001 to 3.000, incl.	80	78 5/	64	77	74 5/	56	7	8 5/	11

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch.
2/ Values are taken in the Longitudinal Transverse Direction (LT).
3/ If the contract or purchase order requires plate in the ordered thickness of 0.500 to 1.000, incl. the minimum mechanical properties shall be listed in the contract or purchase order (see 6.2).
4/ Not Applicable
5/ Ordered Thickness is between 2.500 and 3.000 inches

TABLE IIc. Minimum mechanical properties for class III, AA 7075. 1/, 2/

Thickness, inches	Tensile Strength, ksi	Yield Strength, 0.2% Offset, ksi	Elongation percent
1.000 to 1.500, incl.	70	60	9

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch.
2/ Values are taken in the Longitudinal Transverse Direction (LT).

TABLE IId. Minimum mechanical properties for class IV, AA7181. 1/

Thickness, inches	Tensile Strength, ksi	Yield Strength, 0.2% Offset, ksi	Elongation percent
1.501 to 2.000, incl.	79	73	10
2.001 to 3.000, incl.	78	72	9

1/ Values are taken in the Longitudinal Transverse Direction (LT).

3.4 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.6 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

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3.6.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness, as specified in table III.

3.7 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall contain no more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		**SPECIFIED WIDTH (Inches)**								
		OVER	**0.00**	**39.38**	**59.07**	**78.75**	**98.44**	**118.12**	**137.81**	**157.49**
		THRU	**39.37**	**59.06**	**78.74**	**98.43**	**118.11**	**137.80**	**157.48**	**177.17**
OVER	**THRU**	**TOLERANCES - INCHES (PLUS and MINUS)**								
0.500	**1.000**		0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.001	**1.575**		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.576	**2.362**		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.363	**3.000**		0.075	0.075	0.085	0.100	0.105	0.125	---	---

3.7.1 Ballistic test plates. In addition to the markings in 3.7, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.8 Ballistic test plate information. For each lot of aluminum alloy armor a properly completed Aluminum Armor Test Data Form (see figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.9 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use.

3.9.1 Imperfections. The depth of surface scratches, nicks, pits or gouges shall not reduce the plate thickness below the allowable minimum thickness. Isolated individual pits over 0.015 inches deep but not over 0.030 inches deep and not within 6 inches of each other and do not violate the minimum allowable thickness, are acceptable.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Production inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate of the same alloy; same source of molten metal representing the ingots poured and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One (1) tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two (2) plates, 12 inches by 36 inches of each thickness to be supplied on the contract or purchase order, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer.

4.5.2 Conformance inspection.

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4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one (1) sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one (1) sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

4.5.2.3 Ballistic testing. One (1) plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.3.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.7) and workmanship (see 3.9) requirements.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.6 in accordance with the sampling procedures approved by the procuring activity (see 6.2).

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR									
SPECIFICATION: MIL-DTL-32375		**REVISION: B**					**AMENDMENT: 3**		
FIRING RECORD:		**DATE:**		**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER			**PRIME CONTRACTOR**					
Name:									
Address:									
POC:									
Phone No:									
Fax No:									
TEST ITEM IDENTIFICATION:				Ordered Thickness:					
Lot No. _____ **Plate No.** _____		**Alloy:** _____ **Temper :** _____		**Class:** _____ **Type:** _____					
PURPOSE: ☐ Acceptance ☐ First Article ☐ Development ☐ Reduced Testing ☐ Audit Testing									
SAMPLE: ☐ Primary ☐ Retest (Firing Record No. of Failed Sample)									
CHEMISTRY ANALYSIS	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:									
UTS (ksi):			0.2% YS (ksi):			Elongation (%):			
WELDABILITY:	Thickness	Required Velocity	Actual Velocity	Pass/Fail					
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES** ☐ **NO** ☐									
Test:	**Obl. (deg)**	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**				
Projectile:					
	NOTES:				
SPALL:	Spall Size:	Two Projectile Calibers:	Pass / Fail:		
Lot [☐ met] [☐ failed to meet] the ballistic requirements of specification MIL-DTL-32375					
Government Representative:	Date:	Supplier Representative:	Date:		

FIGURE 1. Aluminum Armor Test Data Form.

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4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E3061, E716, and E1251. In case of any dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal transverse (LT) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	CLASS I & II TYPE A	CLASS I & II TYPE B	CLASS III	CLASS IV
0.500 - 0.749	Cal. .30 AP M2	30	Table A-I	Table A-I	N/A	N/A
0.750 - 0.950	Cal. .50 FSP	0	Table A-II	Table A-II	N/A	N/A
0.951 - 1.500 class I&II 1.000 - 1.500 class III	20-mm FSP	0	Table A-III	Table A-III	Table A-III	N/A
0.750 - 1.500 class I&II 1.000 - 1.500 class III	Cal. .30 AP M2	0	Table A-IV	Table A-IV	Table A-VII	N/A
1.501 - 2.499	Cal. 50 AP M2	0	Table A-V	Table A-V	N/A	Table A-V
2.500 - 3.000	Cal. 50 AP M2	0	N/A	Table A-V	N/A	N/A
2.500 - 3.000	14.5-mm BS-41	0	Table A-VI	N/A	N/A	Table A-VI

4.7.3.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

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4.7.3.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, and 4.8.2.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.9 Reduced testing. At the discretion of the procuring activity (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The armor specified herein as type A is intended for lightweight applications where resistance to ball and armor piercing types of ammunition and multiple hit capabilities are required. The armor specified herein as type B is intended for use on combat and tactical vehicles to protect the occupants from underbody threats (blast). Class II, type A1 is intended to be used on combat vehicles with a thickness of 2.500 to 3.000 inches. Class III armor is intended for use as an applique (B-Kit) for protection against Improvised Explosive Devices (IEDs) and armor piercing (AP) threats. Class IV armor is intended for use as an equivalent to class I and II, type A and is intended for lightweight applications where resistance to ball and armor piercing types of ammunition and multiple hit capabilities are required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify choice of alloy, and ordered thickness (see 1.1)
(c) Specify classification (see 1.2).
(d) When first article is required (see 3.1).
(e) Special mechanical properties and ballistic requirements, if required (see 3.3 and table 2b, footnote 3).
(f) Dimension and tolerance requirements if other than in 3.6 and 3.6.1.
(g) If markings are different (see 3.7).
(h) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(i) The orientation of the ballistic plate is different (see 4.5.2.3).
(j) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(k) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1)
(l) Rejection and retest requirement, if other than in 4.8.
(m) If reduced testing is allowed (see 4.9).
(n) Packaging requirements (see 5.1).

6.3 Density. The density of the specified alloys listed in this specification are as follows:

AA7056 = 0.104 lb/in^3 (2.87 g/cc)
AA7085 = 0.103 lb/in^3 (2.85 g/cc)
AA7075 = 0.101 lb/in^3 (2.81 g/cc)
AA7181 = 0.102 lb/in^3 (2.84 g/cc)

6.4 Tempers. The following tempers have shown to produce acceptable material when used to obtain the desired condition; maximum resistance to armor piercing (AP) projectiles, or maximum resistance to blast.

For class I, type A - temper of T711
For class I, type B - temper of T721

For class II, type A - temper of T761
For class II, type A1 - temper of T741 (for special applications)
For class II, type B - temper of T721

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6.5 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)

6.6 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
Military vehicles
20-mm FSP

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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BALLISTIC TESTING OF ARMOR PLATE, ALUMINUM ALLOY, 7085, 7056, 7075, and 7181, UNWELDABLE APPLIQUE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, un-weldable, 7085, 7056, 7075, and 7181 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VII.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the

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high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification including the plates utilized for acceptance retests are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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TABLE A-I. Minimum required ballistic limits for class I & II - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B		Type A	Type B
0.475	1580	1481	0.580	1822	1679	0.685	2035	1856
0.480	1592	1491	0.585	1832	1688	0.690	2044	1864
0.485	1604	1501	0.590	1843	1697	0.695	2054	1872
0.490	1617	1511	0.595	1854	1705	0.700	2063	1880
0.495	1629	1520	0.600	1864	1714	0.705	2073	1888
0.500 [1/]	1641	1530	0.605	1875	1723	0.710	2082	1896
0.505	1653	1540	0.610	1885	1731	0.715	2092	1904
0.510	1664	1550	0.615	1895	1740	0.720	2101	1911
0.515	1676	1559	0.620	1906	1748	0.725	2110	1919
0.520	1688	1569	0.625	1916	1757	0.730	2120	1927
0.525	1699	1578	0.630	1926	1765	0.735	2129	1935
0.530	1711	1588	0.635	1936	1774	0.740	2138	1942
0.535	1722	1597	0.640	1946	1782	0.745	2147	1950
0.540	1733	1606	0.645	1956	1791	0.749 [2/]	2154	1956
0.545	1745	1616	0.650	1966	1799	0.755	2165	1965
0.550	1756	1625	0.655	1976	1807	0.760	2174	1973
0.555	1767	1634	0.660	1986	1815	0.765	2183	1980
0.560	1778	1643	0.665	1996	1824	0.770	2192	1988
0.565	1789	1652	0.670	2006	1832	0.775	2201	1995
0.570	1800	1661	0.675	2015	1840	0.780	2210	2003
0.575	1811	1670	0.680	2025	1848	0.785	2219	2010

[1/] Specification (type A & B) requirements begin for this ordered thickness.
[2/] Specification (type A & B) requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-II. Minimum required ballistic limits for class I & II - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B		Type A	Type B
0.730	1916	1858	0.815	2287	2191	0.900	2730	2583
0.735	1936	1876	0.820	2311	2212	0.905	2759	2608
0.740	1956	1894	0.825	2335	2234	0.910	2787	2634
0.745	1976	1913	0.830	2359	2255	0.915	2817	2659
0.750 [1/]	1997	1931	0.835	2384	2277	0.920	2846	2685
0.755	2018	1950	0.840	2409	2299	0.925	2876	2711
0.760	2039	1969	0.845	2434	2322	0.930	2906	2738
0.765	2061	1988	0.850	2460	2344	0.935	2936	2764
0.770	2082	2008	0.855	2486	2367	0.940	2967	2791
0.775	2104	2027	0.860	2512	2390	0.945	2998	2819
0.780	2126	2047	0.865	2538	2414	0.950 [2/]	3030	2846
0.785	2148	2067	0.870	2565	2437	0.955	3061	2874
0.790	2171	2087	0.875	2591	2461	0.960	3093	2902
0.795	2193	2107	0.880	2619	2485	0.965	3126	2930
0.800	2216	2128	0.885	2646	2509	0.970	3159	2959
0.805	2240	2149	0.890	2674	2533	0.975	3192	2987
0.810	2263	2169	0.895	2702	2558	0.980	3225	3016

[1/] Specification (type A & B) requirements begin for this ordered thickness.
[2/] Specification (type A & B) requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-III. Minimum required ballistic limits for class I, II, & III - 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II Type A Class III	Class I & II Type B		Class I & II Type A Class III	Class I & II Type B		Class I & II Type A Class III	Class I & II Type B
0.930	1198	1255	1.130	1601	1622	1.330	2140	2097
0.935	1207	1263	1.135	1613	1633	1.335	2155	2111
0.940	1215	1271	1.140	1624	1643	1.340	2171	2124
0.945	1224	1279	1.145	1636	1654	1.345	2187	2138
0.951 [1/]	1235	1289	1.150	1648	1664	1.350	2203	2152
0.955	1242	1296	1.155	1660	1675	1.355	2219	2166
0.960	1251	1304	1.160	1672	1686	1.360	2235	2179
0.965	1260	1313	1.165	1684	1697	1.365	2251	2194
0.970	1269	1321	1.170	1696	1708	1.370	2267	2208
0.975	1279	1330	1.175	1709	1719	1.375	2284	2222
0.980	1288	1338	1.180	1721	1730	1.380	2300	2236
0.985	1297	1347	1.185	1734	1741	1.385	2317	2251
0.990	1307	1355	1.190	1746	1752	1.390	2334	2265
0.995	1316	1364	1.195	1759	1763	1.395	2351	2280
1.000 [2/]	1326	1373	1.200	1772	1775	1.400	2368	2294
1.005	1335	1382	1.205	1785	1786	1.405	2385	2309
1.010	1345	1391	1.210	1798	1798	1.410	2403	2324
1.015	1355	1400	1.215	1811	1809	1.415	2420	2339
1.020	1365	1409	1.220	1824	1821	1.420	2438	2354
1.025	1375	1418	1.225	1837	1833	1.425	2456	2369
1.030	1385	1427	1.230	1851	1844	1.430	2473	2384
1.035	1395	1436	1.235	1864	1856	1.435	2491	2400
1.040	1405	1445	1.240	1878	1868	1.440	2510	2415
1.045	1415	1455	1.245	1891	1880	1.445	2528	2431
1.050	1425	1464	1.250	1905	1892	1.450	2546	2446
1.055	1436	1473	1.255	1919	1905	1.455	2565	2462
1.060	1446	1483	1.260	1933	1917	1.460	2583	2478
1.065	1457	1492	1.265	1947	1929	1.465	2602	2494
1.070	1467	1502	1.270	1961	1942	1.470	2621	2510
1.075	1478	1512	1.275	1976	1954	1.475	2640	2526
1.080	1489	1521	1.280	1990	1967	1.480	2660	2542
1.085	1500	1531	1.285	2004	1979	1.485	2679	2559
1.090	1511	1541	1.290	2019	1992	1.490	2698	2575
1.095	1522	1551	1.295	2034	2005	1.495	2718	2592
1.100	1533	1561	1.300	2048	2018	1.500 [3/]	2738	2609
1.105	1544	1571	1.305	2063	2031	1.505	2758	2625
1.110	1555	1581	1.310	2078	2044	1.510	2778	2642
1.115	1566	1591	1.315	2094	2057	1.515	2798	2659
1.120	1578	1602	1.320	2109	2070	1.520	2818	2676
1.125	1589	1612	1.325	2124	2084	1.525	2839	2694

[1/] Specification (class I & II; type A & B) requirements begin for this ordered thickness.
[2/] Specification (class III) requirements begin for this ordered thickness.
[3/] Specification (class I & II; type A & B and class III) requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-IV. Minimum required ballistic limits for class I & II - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B		Type A	Type B
0.730	1912	1787	0.910	2192	2040	1.090	2440	2265
0.735	1920	1794	0.915	2199	2046	1.095	2446	2271
0.740	1928	1802	0.920	2206	2053	1.100	2453	2277
0.745	1937	1809	0.925	2214	2059	1.105	2459	2282
0.750 1/	1945	1816	0.930	2221	2066	1.110	2466	2288
0.755	1953	1824	0.935	2228	2072	1.115	2472	2294
0.760	1961	1831	0.940	2235	2079	1.120	2479	2300
0.765	1969	1839	0.945	2242	2085	1.125	2485	2306
0.770	1977	1846	0.950	2249	2092	1.130	2492	2312
0.775	1985	1853	0.955	2256	2098	1.135	2498	2318
0.780	1994	1860	0.960	2263	2105	1.140	2504	2323
0.785	2002	1868	0.965	2270	2111	1.145	2511	2329
0.790	2009	1875	0.970	2278	2117	1.150	2517	2335
0.795	2017	1882	0.975	2284	2124	1.155	2523	2341
0.800	2025	1889	0.980	2291	2130	1.160	2530	2346
0.805	2033	1896	0.985	2298	2136	1.165	2536	2352
0.810	2041	1903	0.990	2305	2143	1.170	2542	2358
0.815	2049	1910	0.995	2312	2149	1.175	2549	2364
0.820	2057	1917	1.000	2319	2155	1.180	2555	2369
0.825	2064	1924	1.005	2326	2161	1.185	2561	2375
0.830	2072	1931	1.010	2333	2168	1.190	2567	2381
0.835	2080	1938	1.015	2340	2174	1.195	2574	2386
0.840	2087	1945	1.020	2347	2180	1.200	2580	2392
0.845	2095	1952	1.025	2353	2186	1.205	2586	2397
0.850	2103	1959	1.030	2360	2192	1.210	2592	2403
0.855	2110	1966	1.035	2367	2198	1.215	2598	2409
0.860	2118	1973	1.040	2374	2205	1.220	2604	2414
0.865	2125	1979	1.045	2380	2211	1.225	2611	2420
0.870	2133	1986	1.050	2387	2217	1.230	2617	2425
0.875	2140	1993	1.055	2394	2223	1.235	2623	2431
0.880	2148	2000	1.060	2400	2229	1.240	2629	2436
0.885	2155	2006	1.065	2407	2235	1.245	2635	2442
0.890	2162	2013	1.070	2414	2241	1.250	2641	2447
0.895	2170	2020	1.075	2420	2247	1.255	2647	2453
0.900	2177	2026	1.080	2427	2253	1.260	2653	2458
0.905	2185	2033	1.085	2433	2259	1.265	2659	2464

1/ Specification (type A & B) requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-IV. Minimum required ballistic limits for class I & II - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B		Type A	Type B
1.270	2665	2469	1.355	2765	2560	1.440	2861	2648
1.275	2671	2475	1.360	2771	2566	1.445	2867	2653
1.280	2677	2480	1.365	2776	2571	1.450	2872	2658
1.285	2683	2486	1.370	2782	2576	1.455	2878	2663
1.290	2689	2491	1.375	2788	2581	1.460	2884	2668
1.295	2695	2496	1.380	2794	2586	1.465	2889	2673
1.300	2701	2502	1.385	2799	2592	1.470	2895	2678
1.305	2707	2507	1.390	2805	2597	1.475	2900	2683
1.310	2712	2513	1.395	2811	2602	1.480	2906	2688
1.315	2718	2518	1.400	2816	2607	1.485	2911	2693
1.320	2724	2523	1.405	2822	2612	1.490	2917	2698
1.325	2730	2529	1.410	2828	2617	1.495	2922	2703
1.330	2736	2534	1.415	2833	2623	1.500 [2/]	2928	2708
1.335	2742	2539	1.420	2839	2628	1.505	2933	2713
1.340	2748	2544	1.425	2845	2633	1.510	2938	2718
1.345	2753	2550	1.430	2850	2638	1.515	2944	2723
1.350	2759	2555	1.435	2856	2643	1.520	2949	2728

[2/] Specification (type A & B) requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits for class I, II, & IV - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II Type A Class IV	Class I & II Type B		Class I & II Type A Class IV	Class I & II Type B		Class I & II Type A Class IV	Class I & II Type B
1.480	2080	1907	1.670	2258	2081	1.860	2423	2241
1.485	2085	1912	1.675	2262	2085	1.865	2427	2245
1.490	2090	1917	1.680	2267	2089	1.870	2431	2249
1.495	2095	1921	1.685	2271	2094	1.875	2435	2253
1.501 1/	2100	1927	1.690	2276	2098	1.880	2439	2257
1.505	2104	1931	1.695	2280	2102	1.885	2444	2261
1.510	2109	1936	1.700	2285	2107	1.890	2448	2265
1.515	2114	1940	1.705	2289	2111	1.895	2452	2269
1.520	2119	1945	1.710	2294	2115	1.900	2456	2273
1.525	2124	1950	1.715	2298	2120	1.905	2460	2277
1.530	2128	1954	1.720	2302	2124	1.910	2464	2281
1.535	2133	1959	1.725	2307	2128	1.915	2468	2285
1.540	2138	1964	1.730	2311	2132	1.920	2472	2289
1.545	2143	1968	1.735	2316	2137	1.925	2477	2293
1.550	2147	1973	1.740	2320	2141	1.930	2481	2297
1.555	2152	1977	1.745	2324	2145	1.935	2485	2301
1.560	2157	1982	1.750	2329	2149	1.940	2489	2305
1.565	2161	1987	1.755	2333	2154	1.945	2493	2309
1.570	2166	1991	1.760	2337	2158	1.950	2497	2313
1.575	2171	1996	1.765	2342	2162	1.955	2501	2317
1.580	2175	2000	1.770	2346	2166	1.960	2505	2320
1.585	2180	2005	1.775	2350	2170	1.965	2509	2324
1.590	2185	2009	1.780	2355	2175	1.970	2513	2328
1.595	2189	2014	1.785	2359	2179	1.975	2517	2332
1.600	2194	2018	1.790	2363	2183	1.980	2521	2336
1.605	2199	2023	1.795	2368	2187	1.985	2525	2340
1.610	2203	2027	1.800	2372	2191	1.990	2529	2344
1.615	2208	2032	1.805	2376	2195	1.995	2533	2348
1.620	2213	2036	1.810	2380	2200	2.000	2537	2352
1.625	2217	2041	1.815	2385	2204	2.005	2541	2355
1.630	2222	2045	1.820	2389	2208	2.010	2545	2359
1.635	2226	2050	1.825	2393	2212	2.015	2549	2363
1.640	2231	2054	1.830	2397	2216	2.020	2553	2367
1.645	2235	2059	1.835	2402	2220	2.025	2557	2371
1.650	2240	2063	1.840	2406	2224	2.030	2561	2375
1.655	2244	2067	1.845	2410	2228	2.035	2565	2379
1.660	2249	2072	1.850	2414	2232	2.040	2569	2382
1.665	2253	2076	1.855	2419	2237	2.045	2573	2386

1/ Specification class I & class II, type A & B & class IV requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits for class I, II, & IV - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II Type A Class IV	Class I & II Type B		Class I & II Type A Class IV	Class I & II Type B		Class I & II Type A Class IV	Class I & II Type B
2.050	2577	2390	2.225	2711	2520	2.400	2839	2643
2.055	2581	2394	2.230	2715	2523	2.405	2843	2647
2.060	2585	2398	2.235	2719	2527	2.410	2847	2650
2.065	2589	2401	2.240	2723	2531	2.415	2850	2653
2.070	2593	2405	2.245	2726	2534	2.420	2854	2657
2.075	2597	2409	2.250	2730	2538	2.425	2857	2660
2.080	2601	2413	2.255	2734	2541	2.430	2861	2664
2.085	2604	2416	2.260	2737	2545	2.435	2864	2667
2.090	2608	2420	2.265	2741	2548	2.440	2868	2671
2.095	2612	2424	2.270	2745	2552	2.445	2871	2674
2.100	2616	2428	2.275	2749	2556	2.450	2875	2677
2.105	2620	2432	2.280	2752	2559	2.455	2878	2681
2.110	2624	2435	2.285	2756	2563	2.460	2882	2684
2.115	2628	2439	2.290	2760	2566	2.465	2885	2687
2.120	2632	2443	2.295	2763	2570	2.470	2889	2691
2.125	2635	2446	2.300	2767	2573	2.475	2892	2694
2.130	2639	2450	2.305	2771	2577	2.480	2896	2698
2.135	2643	2454	2.310	2774	2580	2.485	2899	2701
2.140	2647	2458	2.315	2778	2584	2.490	2903	2704
2.145	2651	2461	2.320	2782	2587	2.495	2906	2708
2.150	2655	2465	2.325	2785	2591	2.499 2/	2909	2710
2.155	2658	2469	2.330	2789	2594	2.505	2913	2714
2.160	2662	2472	2.335	2793	2598	2.510	2917	2718
2.165	2666	2476	2.340	2796	2601	2.515	2920	2721
2.170	2670	2480	2.345	2800	2605	2.520	2924	2724
2.175	2674	2483	2.350	2803	2608	2.525	N/A	2728
2.180	2677	2487	2.355	2807	2612	2.530	N/A	2731
2.185	2681	2491	2.360	2811	2615	2.535	N/A	2734
2.190	2685	2494	2.365	2814	2619	2.540	N/A	2738
2.195	2689	2498	2.370	2818	2622	2.545	N/A	2741
2.200	2693	2502	2.375	2821	2626	2.550	N/A	2744
2.205	2696	2505	2.380	2825	2629	2.555	N/A	2748
2.210	2700	2509	2.385	2829	2633	2.560	N/A	2751
2.215	2704	2512	2.390	2832	2636	2.565	N/A	2754
2.220	2708	2516	2.395	2836	2640	2.570	N/A	2758

2/ Specification class I & class II, type A & class IV requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits for class I & II; type B only - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps Type B	Thickness, inches	Required BL(P), fps Type B	Thickness, inches	Required BL(P), fps Type B
2.575	2761	2.730	2861	2.885	2958
2.580	2764	2.735	2865	2.890	2961
2.585	2768	2.740	2868	2.895	2964
2.590	2771	2.745	2871	2.900	2967
2.595	2774	2.750	2874	2.905	2971
2.600	2777	2.755	2877	2.910	2974
2.605	2781	2.760	2880	2.915	2977
2.610	2784	2.765	2884	2.920	2980
2.615	2787	2.770	2887	2.925	2983
2.620	2790	2.775	2890	2.930	2986
2.625	2794	2.780	2893	2.935	2989
2.630	2797	2.785	2896	2.940	2992
2.635	2800	2.790	2899	2.945	2995
2.640	2803	2.795	2902	2.950	2998
2.645	2807	2.800	2906	2.955	3001
2.650	2810	2.805	2909	2.960	3004
2.655	2813	2.810	2912	2.965	3007
2.660	2816	2.815	2915	2.970	3010
2.665	2820	2.820	2918	2.975	3013
2.670	2823	2.825	2921	2.980	3016
2.675	2826	2.830	2924	2.985	3019
2.680	2829	2.835	2927	2.990	3022
2.685	2833	2.840	2930	2.995	3025
2.690	2836	2.845	2934	3.000 [3/]	3028
2.695	2839	2.850	2937	3.005	3031
2.700	2842	2.855	2940	3.010	3034
2.705	2845	2.860	2943	3.015	3037
2.710	2849	2.865	2946	3.020	3040
2.715	2852	2.870	2949	3.025	3043
2.720	2855	2.875	2952	3.030	3046
2.725	2858	2.880	2955	3.035	3049

[3/] Type B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VI. Minimum required ballistic limits for class I & II; type A and A1 & class IV 14.5-mm BS-41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps Class I & II Type A & A1 Class IV	Thickness, inches	Required BL(P), fps Class I & II Type A & A1 Class IV	Thickness, inches	Required BL(P), fps Class I & II Type A & A1 Class IV
2.480	2650	2.665	2770	2.850	2886
2.485	2653	2.670	2774	2.855	2889
2.490	2656	2.675	2777	2.860	2892
2.495	2660	2.680	2780	2.865	2895
2.500 [1/]	2664	2.685	2783	2.870	2898
2.505	2666	2.690	2786	2.875	2901
2.510	2670	2.695	2789	2.880	2904
2.515	2673	2.700	2793	2.885	2907
2.520	2676	2.705	2796	2.890	2910
2.525	2680	2.710	2799	2.895	2914
2.530	2683	2.715	2802	2.900	2917
2.535	2686	2.720	2805	2.905	2920
2.540	2689	2.725	2808	2.910	2923
2.545	2693	2.730	2812	2.915	2926
2.550	2696	2.735	2815	2.920	2929
2.555	2699	2.740	2818	2.925	2932
2.560	2703	2.745	2821	2.930	2935
2.565	2706	2.750	2824	2.935	2938
2.570	2709	2.755	2827	2.940	2941
2.575	2712	2.760	2830	2.945	2944
2.580	2716	2.765	2834	2.950	2947
2.585	2719	2.770	2837	2.955	2950
2.590	2722	2.775	2840	2.960	2953
2.595	2725	2.780	2843	2.965	2956
2.600	2729	2.785	2846	2.970	2959
2.605	2732	2.790	2849	2.975	2962
2.610	2735	2.795	2852	2.980	2965
2.615	2738	2.800	2855	2.985	2968
2.620	2742	2.805	2858	2.990	2971
2.625	2745	2.810	2861	2.995	2974
2.630	2748	2.815	2865	3.000 [2]	2977
2.635	2751	2.820	2868	3.005	2980
2.640	2754	2.825	2871	3.010	2982
2.645	2758	2.830	2874	3.015	2985
2.650	2761	2.835	2877	3.020	2988
2.655	2764	2.840	2880	3.025	2991
2.660	2767	2.845	2883	3.030	2994

[1/] class I & II, type A, class II type A1, & class IV requirements begin for this ordered thickness.
[2/] class I & II, type A, class II type A1, & class IV requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VII. Minimum required ballistic limits for class III - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps Class III	Thickness, inches	Required BL(P), fps Class III	Thickness, inches	Required BL(P), fps Class III
0.975	2171	1.160	2405	1.345	2639
0.980	2177	1.165	2411	1.350	2645
0.985	2184	1.170	2417	1.355	2652
0.990	2190	1.175	2424	1.360	2658
0.995	2196	1.180	2430	1.365	2664
1.000 1/	2202	1.185	2436	1.370	2671
1.005	2209	1.190	2443	1.375	2677
1.010	2215	1.195	2449	1.380	2683
1.015	2221	1.200	2455	1.385	2690
1.020	2228	1.205	2461	1.390	2696
1.025	2234	1.210	2468	1.395	2702
1.030	2240	1.215	2474	1.400	2709
1.035	2247	1.220	2480	1.405	2715
1.040	2253	1.225	2487	1.410	2721
1.045	2259	1.230	2493	1.415	2728
1.050	2266	1.235	2499	1.420	2734
1.055	2272	1.240	2506	1.425	2740
1.060	2278	1.245	2512	1.430	2747
1.065	2285	1.250	2518	1.435	2753
1.070	2291	1.255	2525	1.440	2759
1.075	2297	1.260	2531	1.445	2766
1.080	2303	1.265	2537	1.450	2772
1.085	2310	1.270	2544	1.455	2779
1.090	2316	1.275	2550	1.460	2785
1.095	2322	1.280	2556	1.465	2791
1.100	2329	1.285	2563	1.470	2798
1.105	2335	1.290	2569	1.475	2804
1.110	2341	1.295	2575	1.480	2810
1.115	2348	1.300	2582	1.485	2817
1.120	2354	1.305	2588	1.490	2823
1.125	2360	1.310	2594	1.495	2829
1.130	2367	1.315	2601	1.500 2/	2836
1.135	2373	1.320	2607	1.505	2842
1.140	2379	1.325	2614	1.510	2848
1.145	2386	1.330	2620	1.515	2855
1.150	2392	1.335	2626	1.520	2861
1.155	2398	1.340	2633	1.525	2868

1/ class III requirements begin for this ordered thickness.
2/ class III requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Review activities:
Army - AR, AT, AV, TE
DLA - IS

Preparing activity:
ARMY - MR

Project 9535-2021-002

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